4 A non-uniform L-shaped beam of weight 10 N is attached to a wall using a hinge. The longer part of the beam is 50 cm and the shorter part of the beam is 10 cm. The weight of the beam acts at a point on the beam that is at a horizonal distance of *x* away from the hinge. The beam is held at rest using a cord, such that the longer part of the beam is horizontal. *AO* is horizontal while *OC* is vertical. The tension in the cord is 5.0 N, at an angle of 60° from *OC*, as shown in Fig. 4.1.

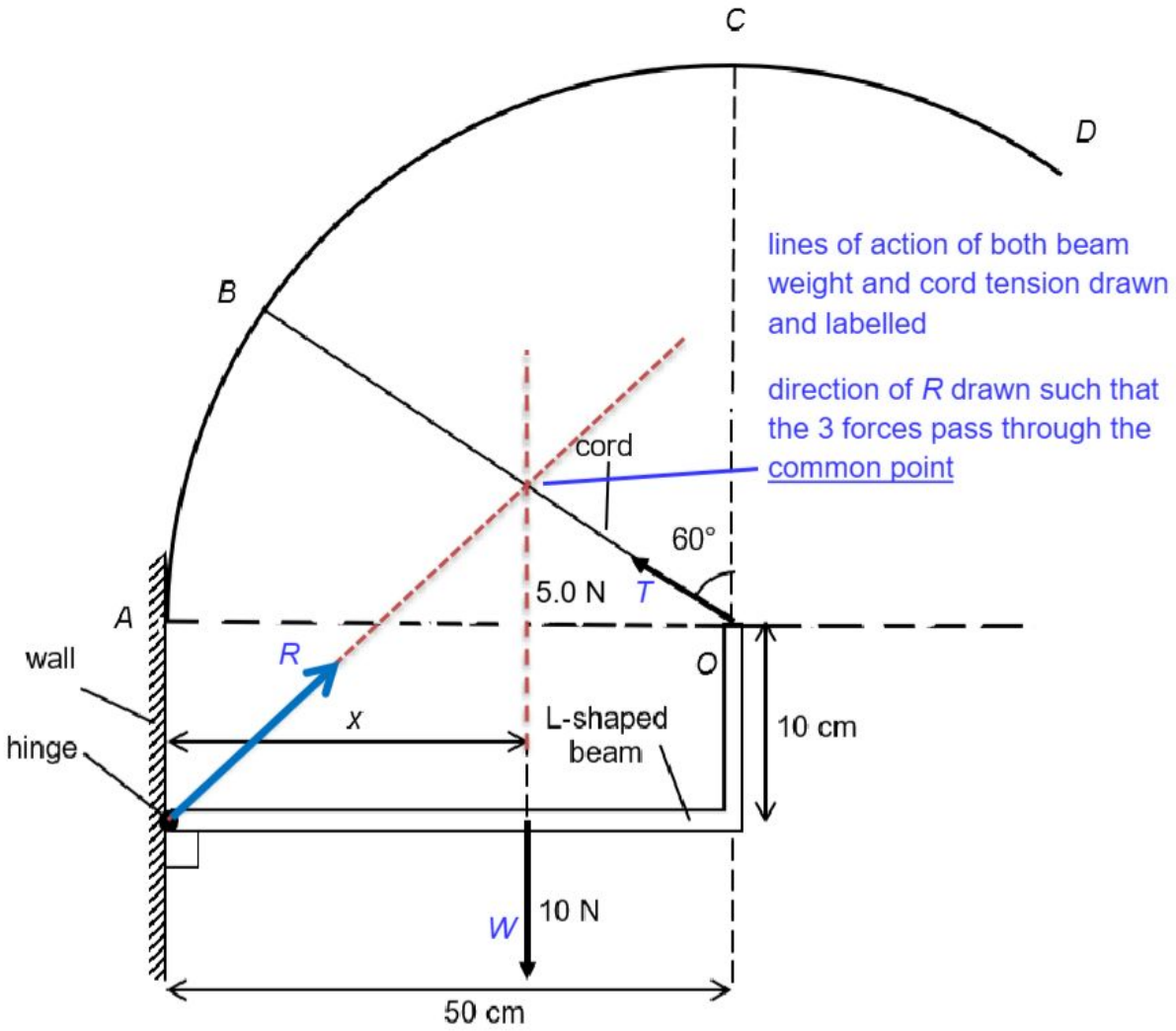


Fig. 4.1

(a) On Fig. 4.1, draw and label the contact force *R* acting on the L-shaped beam due to the hinge.

[2]

(b) By considering the forces acting on the beam, calculate the magnitude of R.

$\sum F_x = 0 \Rightarrow R_x = T_x = 5.0\sin 60^\circ = 4.33$ N
$\sum F_y = 0 \Rightarrow R_y + T_y = W \Rightarrow R_y = W - T_y = 10 - 5.0\cos 60^\circ = 7.50$ N
} [M1]

$|R| = \sqrt{R_x^2 + R_y^2} = \sqrt{4.33^2 + (-7.50)^2}$ [M1]
$= 8.66$ N
$= 8.7$ N (to 2 s.f.) [A1]

magnitude of R = ... N [3]

(c) Determine the value of x.

using hinge as pivot:
clockwise moment due to weight = anticlockwise moment due to tension

$Wx = \left[T\left(\cos 60^\circ\right)\right](0.50) + \left[T\left(\sin 60^\circ\right)\right](0.10)$
$10x = 5.0\left[0.50\left(\cos 60^\circ\right) + 0.10\left(\sin 60^\circ\right)\right]$ [M1]
$x = 0.168$ m $= 0.17$ m $= 17$ cm [A1]

x = ... cm [2]

(d) *ABCD* are different points on a circle where the centre is at point *O*. *C* is at the point vertically above point *O*. The cord moves along the arc of the circle such that the beam remains stationary.

(i) State how the magnitude of the contact force *R* would change as the cord moves from *B* to *C*.

Magnitude *R* decreases as the cord moves from *B* to *C*. [B1]

..... [1]

(ii) State how the direction of the contact force *R* would change as the cord moves from *C* to *D*. Mathematical calculation is not required in this part.

At *C*, *R* acts vertically upwards. As the cord moves from *C* to *D*, the direction of *R* rotates anticlockwise into the wall. [B1]

..... [1]

[Total: 9]